

BrainSight: AI-Powered Brain Tumor Detection

Revolutionizing Diagnostic Radiology

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Subject & Institution:

- Introduction to Innovation, IP Management and Entrepreneurship
- VIT Vellore



The Problem: A Crisis in Timely Diagnosis

→ Slow, Costly, and Error-Prone Diagnosis

Traditional diagnostic methods for brain tumors are resource-intensive, leading to long wait times and potential human error in interpretation.

→ Radiologist Shortage and High Workload

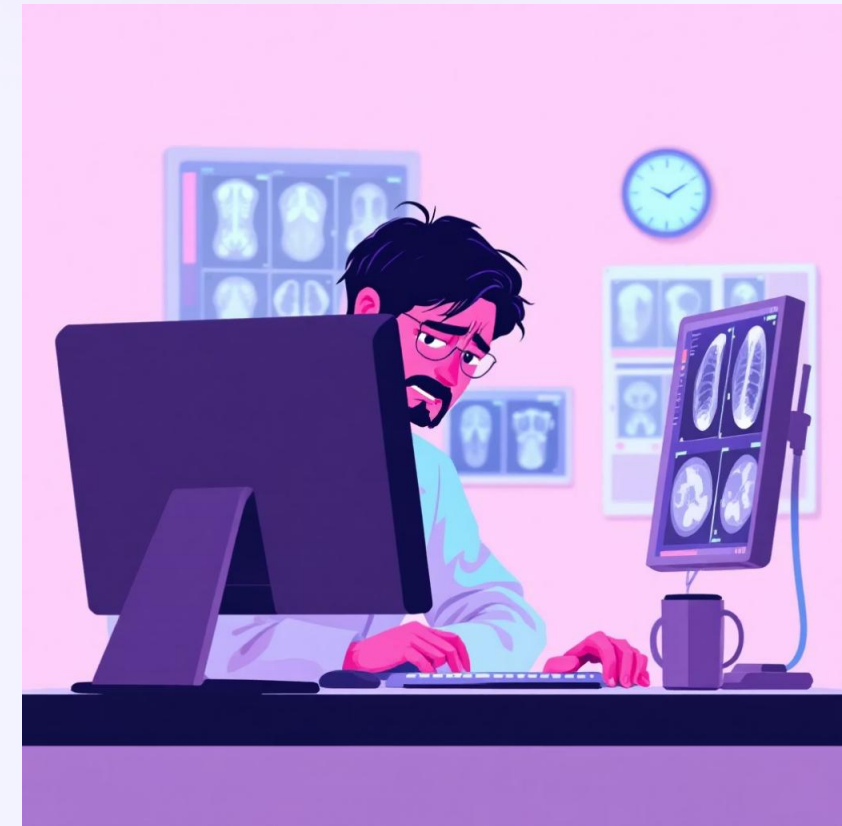
The global deficit of specialized radiologists results in heavy workloads, increasing burnout and delaying critical diagnoses, especially in non-metro areas.

→ Lack of Advanced Tools in Rural Areas

Many rural and tier-2 hospitals lack access to the latest advanced diagnostic infrastructure and expert consultation, hindering quality care.

→ Critical Impact of Delayed Detection

Delayed detection significantly worsens patient prognosis, leading to poor treatment outcomes and higher healthcare costs.





The Market Opportunity

AI-driven solutions are poised to fill a critical gap in medical imaging, offering efficiency and scalability where human resources are constrained.



Rapid Sector Growth

The AI in Healthcare market is projecting robust growth, estimated at a 45% Compound Annual Growth Rate (CAGR) from 2024 to 2030.



Increasing Demand for AI-Aided Radiology

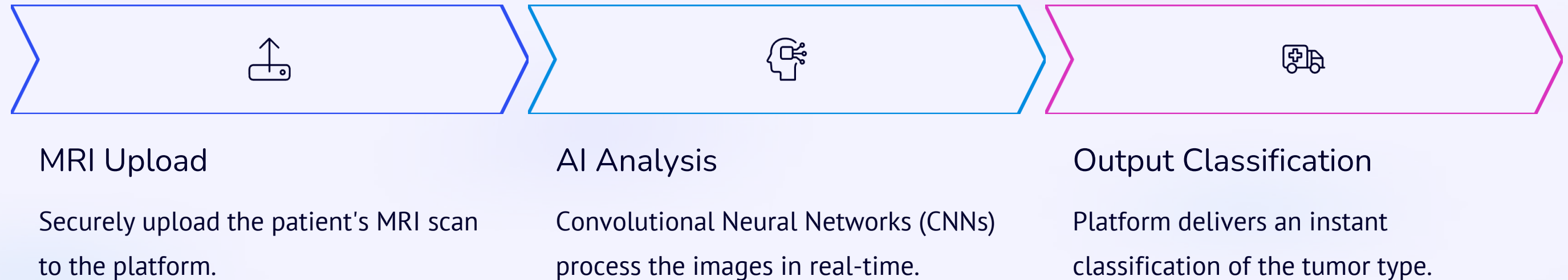
There is a growing global demand for tools that assist radiologists, particularly in high-volume, complex imaging modalities like MRI.



Need for Scalable, Low-Cost Technology

Hospitals, medical colleges, and diagnostic centers are actively seeking affordable, scalable diagnostic technologies to improve patient throughput and accuracy.

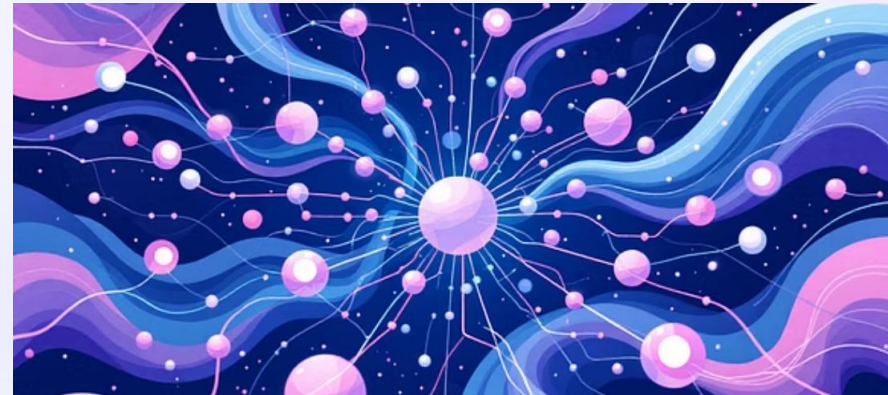
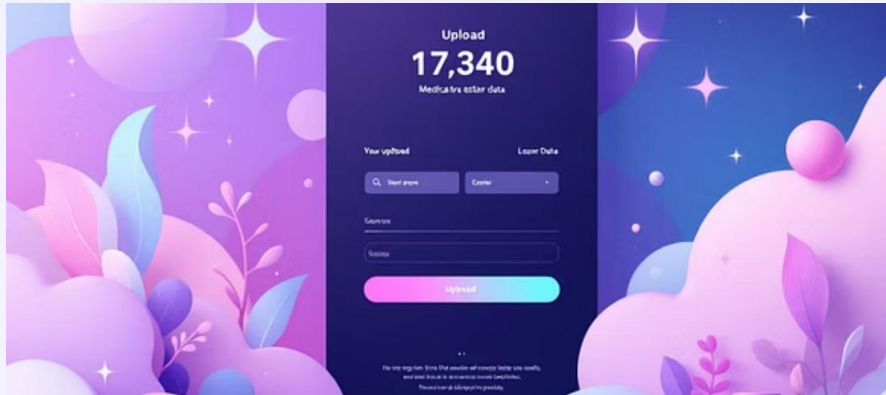
Introducing BrainSight: Our AI Diagnostic Platform



BrainSight automatically detects and classifies brain tumors from MRI scans, providing real-time, accurate, and low-cost medical insights.

 Tumor Types Detected: Glioma, Meningioma, Pituitary Tumor, or No Tumor.

Product Demonstration & Technology



User Interface

Simple, intuitive web application featuring secure upload, immediate results display, and a clear confidence score for physician review.

Deployment & Scalability

Deployed via a robust Flask API and hosted on reliable cloud services (e.g., Render/AWS) ensuring high availability and global scalability.

Core Technology

Proprietary CNN architecture trained on vast MRI datasets using the TensorFlow framework, specifically optimized for brain tumor morphology.

Validated Accuracy

Achieved an independently validated accuracy rate exceeding **95%** on diverse clinical test data.

Unique Value Proposition

BrainSight offers a powerful combination of speed, accuracy, and accessibility unmatched by current high-cost enterprise solutions.



Real-Time Prediction

Diagnosis delivered in under 10 seconds, accelerating patient workflow and critical decision-making.



Exceptional Accuracy

Consistently achieves high diagnostic performance with 90–95% accuracy in clinical environments.



Affordable Solution

Our subscription model is up to 10 times cheaper than competing high-end radiology AI platforms.



Low-Bandwidth Operation

Optimized for seamless operation even in areas with limited or intermittent internet connectivity, ideal for rural adoption.



Explainable AI (XAI)

Provides clear, detailed reports explaining the AI's decision-making process, fostering trust and collaboration with physicians.

Market Segments & Future Scope

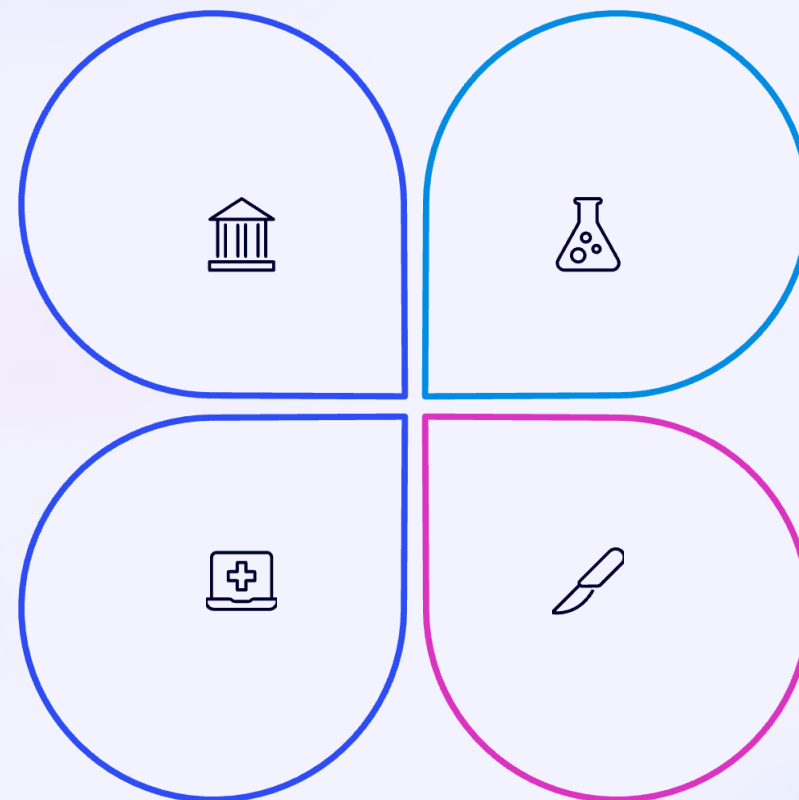
We are targeting core clinical providers while establishing a presence in academic and research sectors, paving the way for future consumer integration.

Primary Market (B2B)

Hospitals, large diagnostic centers, and telemedicine solution providers seeking to improve efficiency.

Telemedicine Firms

Integrating our API directly into their virtual consultation platforms for remote diagnostics.



Secondary Market (A2B)

Medical colleges, universities, and research laboratories licensing the technology for academic use and internal projects.

Future Integration

Potential partnership with health insurance providers and wellness applications for proactive screening services.

Multi-Channel Business Model

A diversified revenue model encompassing B2B subscriptions, academic licensing, and future B2C services.

B2B (Business to Business)

Subscription SaaS

Annual license fees for institutions: Ranging from ₹1 Lakh to ₹5 Lakh per year.

Pay-per-Use API

Transactional revenue for on-demand analysis: Approximately ₹150 per MRI scan analyzed.

Enterprise Partnerships

Custom development and integration projects for large hospital chains.

A2B (Academia to Business)

Research Licensing

Annual license fee for academic use: Starting at ₹50K/year.

R&D Collaboration

Revenue from joint research and development projects with universities.

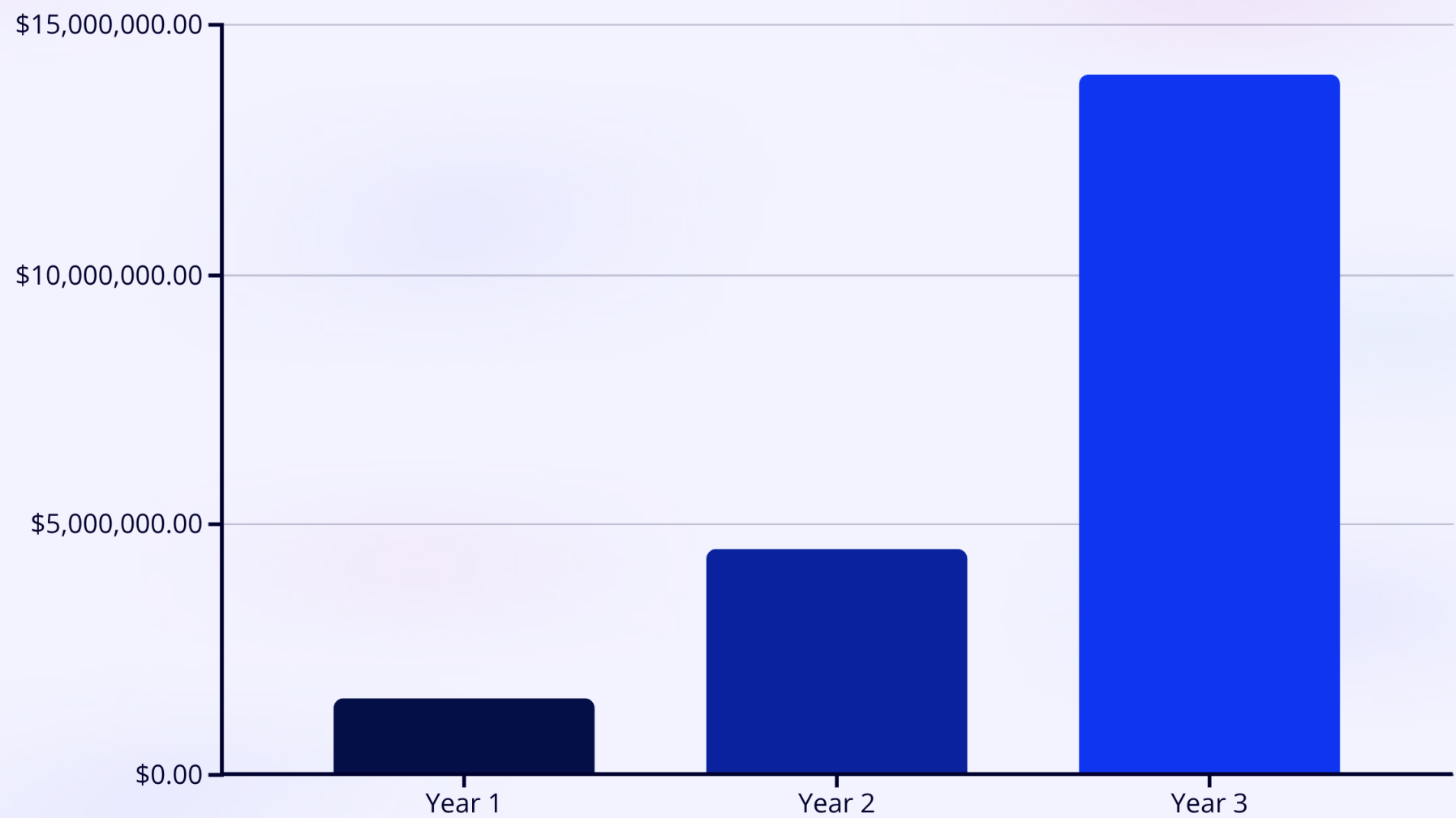
Workshops & Training

Educational revenue stream: ₹1K–₹3K per participant for specialized training.

Future B2C Scope includes personal interpretation services and premium health subscription plans.

Projected Revenue Growth

Based on our scalable SaaS model and low operational costs, we project aggressive revenue growth over the first three years.



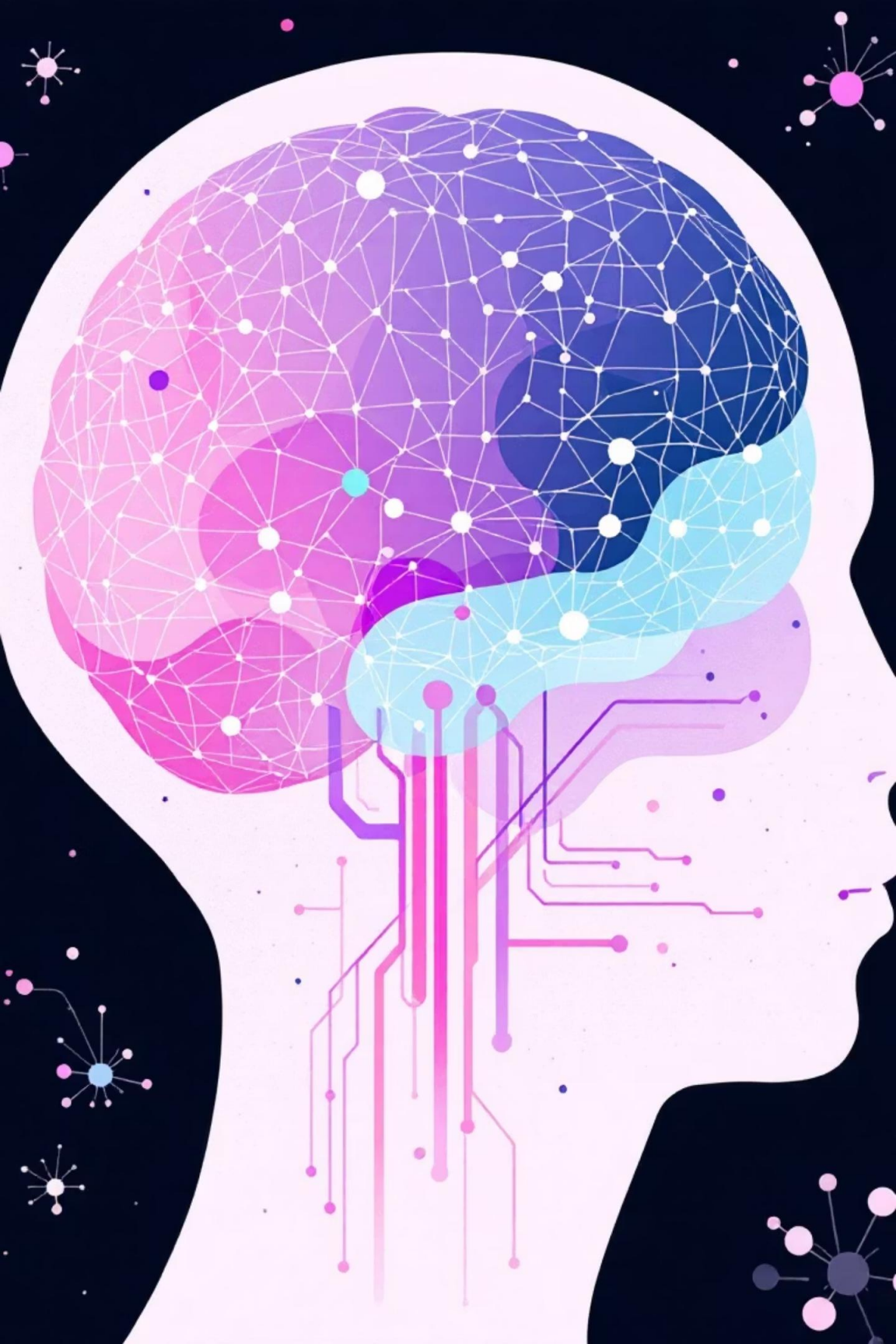
Key Revenue Drivers:

- B2B Subscription uptake from tier-2 city hospitals.
- Increasing utilization of API Pay-per-Use model by smaller clinics.
- Growth in A2B licensing and specialized training revenue.
- Pilot B2C subscriptions starting in Year 3.

Competitive Analysis: Where BrainSight Excels

Feature	Qure.ai	Zebra Medical Vision	BrainSight
Primary Focus	General Radiology	Multimodal Imaging	Brain MRI Specialized
Cost Structure	High Enterprise Cost	Very High Cost	Low-Cost/Affordable SaaS
Deployment Model	Complex Integration	Enterprise	Web-Based Platform
Explainability	Standard	Moderate	High (XAI Reports)

BrainSight is strategically positioned as the cost-effective, specialized solution for brain tumor detection, focusing on accessibility and physician trust.



Technology Stack

BrainSight is built on a robust, enterprise-grade technology foundation designed for accuracy, scalability, and seamless integration into healthcare workflows.



Machine Learning

Convolutional Neural Network (CNN) architecture powered by TensorFlow and Keras for precise diagnostic analysis



Backend Infrastructure

Flask framework in Python ensures reliable processing and rapid response times for clinical workflows



User Interface

Intuitive HTML, CSS, and JavaScript frontend designed for radiologists and clinicians

Data Management

SQLite database provides secure, efficient storage of patient data and diagnostic results

Cloud Deployment

Render and AWS infrastructure enable scalable, reliable access across healthcare networks



Intellectual Property Portfolio

BrainSight's competitive advantage is protected through a comprehensive IP strategy that safeguards our innovation and establishes market leadership.

01

Proprietary CNN Model

Our neural network is trained on carefully curated datasets, creating a unique diagnostic algorithm that competitors cannot replicate

02

Copyrighted Software

Complete protection of source code and web interface ensures exclusive control over our technology platform

03

BrainSight Trademark

Registered trademark establishes brand recognition and prevents market confusion in the neuro-diagnostic space

04

Patent Pipeline

Explainable AI algorithm offers significant patent potential, creating additional barriers to entry for competitors

- ❑ Our multi-layered IP protection creates sustainable competitive advantages and increases enterprise valuation for potential investors and acquisition targets.



Go-to-Market Strategy

Our phased market entry approach combines strategic pilots, targeted marketing, and key partnerships to establish BrainSight as the leading AI diagnostic solution.



Pilot Program

Launch with 3 partner hospitals and 2 medical colleges to validate clinical efficacy and gather testimonials



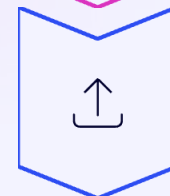
Marketing Outreach

Targeted campaigns via LinkedIn, medical expos, and educational webinars reaching decision-makers



Strategic Partnerships

Collaborate with telehealth platforms and insurance providers to expand market reach



National Scaling

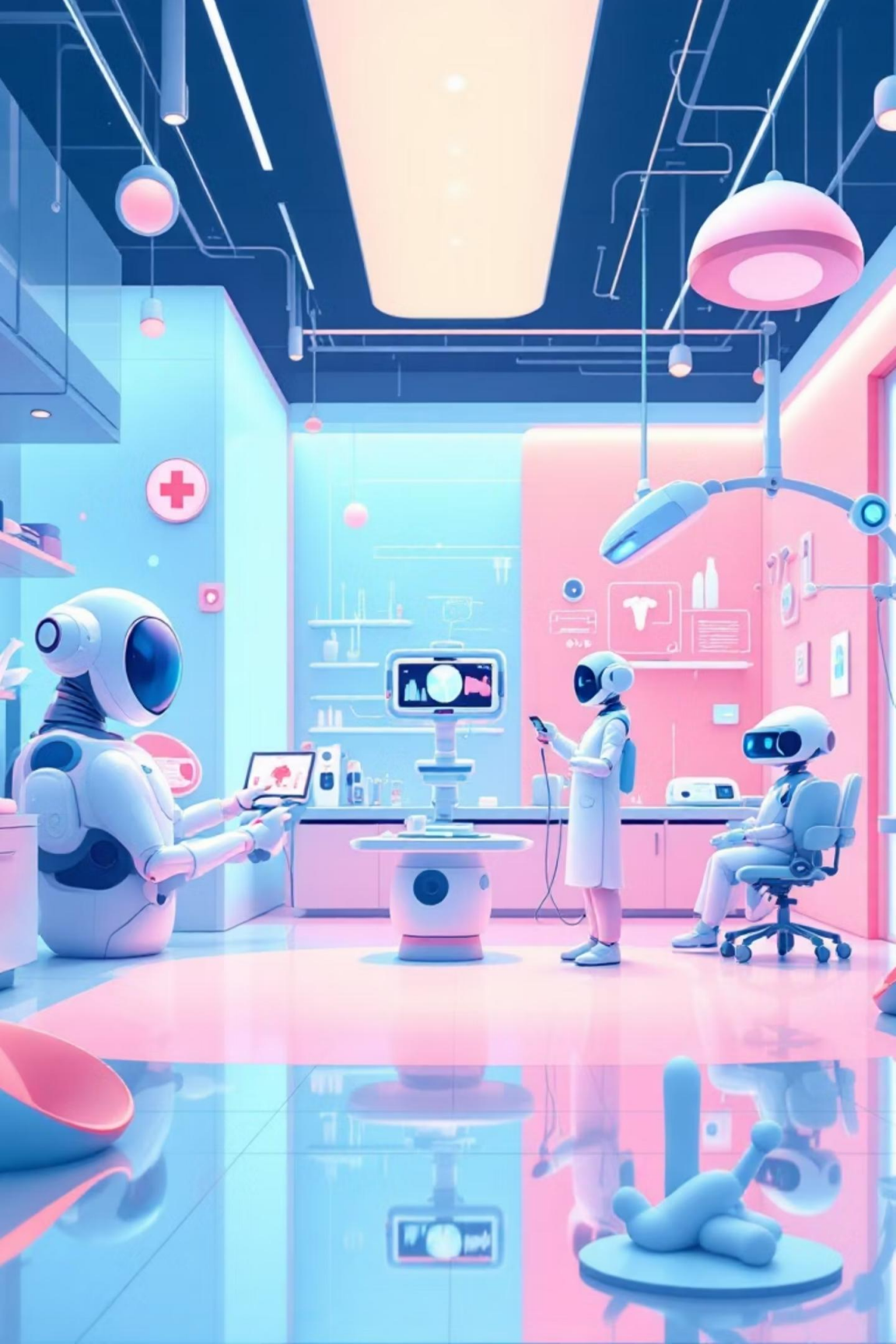
Cloud-based deployment enables rapid expansion across healthcare systems nationwide

Primary Target Markets

- Regional hospital networks
- Academic medical centers
- Radiology groups

Key Success Metrics

- Pilot adoption rates
- Diagnostic accuracy validation
- Time-to-diagnosis reduction



Future Expansion Roadmap

BrainSight's vision extends beyond brain tumor detection. Our technology platform is designed for continuous evolution, expanding into new diagnostic modalities and patient-centered applications.



Multi-Modality Integration

Expand diagnostic capabilities to include CT scans, EEG readings, and PET imaging for comprehensive neurological assessment



Mobile Patient Access

Develop patient-facing mobile application enabling direct access to results, educational resources, and care coordination



Predictive Analytics

Implement advanced modeling for tumor progression forecasting, enabling proactive treatment planning and improved outcomes



EHR System Integration

Seamless connectivity with hospital Electronic Health Record systems for streamlined clinical workflows and data sharing



Each expansion phase builds on proven technology while opening new revenue streams and market opportunities in the rapidly growing AI healthcare sector.



BrainSight

Empowering Early Detection Through AI

We are taking a decisive step toward accessible, affordable, and accurate neuro-diagnostics. BrainSight represents more than technology—it's a commitment to saving lives through earlier intervention and more precise diagnosis.

Accessible

Cloud-based deployment brings advanced AI diagnostics to hospitals of all sizes

Affordable

Reduced diagnostic costs and improved efficiency benefit healthcare systems and patients

Accurate

State-of-the-art CNN technology delivers reliability that clinicians can trust

The future of neuro-diagnostics is here. Join us in transforming brain tumor detection and improving patient outcomes across America.